

MAP & Conjugate Priors

MLE, with a prior's vote

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Mathematical Foundations for AI · IIT Gandhinagar

Three heads in three flips

Where Lecture 14 left us

L14 fixed the data, varied the parameter, and maximized the likelihood.

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} P(D \mid \theta)$$

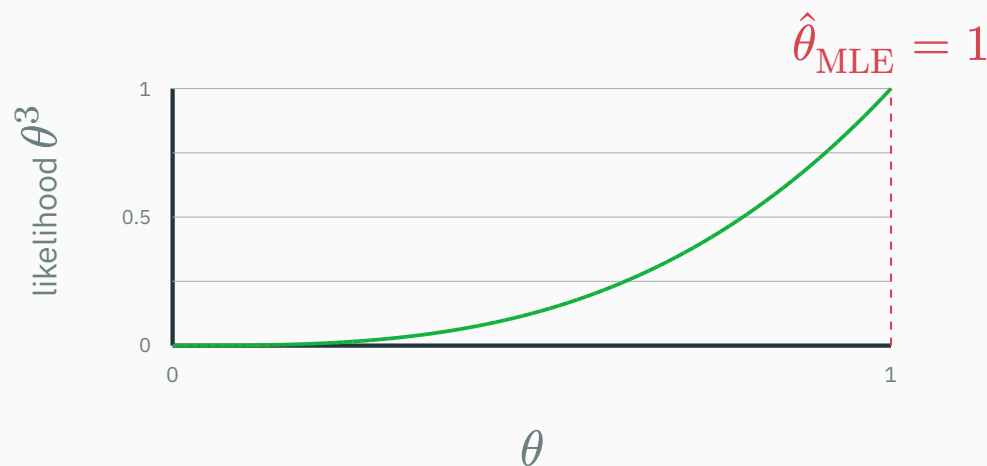
- The coin: n_H heads in n flips $\Rightarrow \hat{\theta}_{\text{MLE}} = n_H/n$ — count and divide.
- With fixed-variance Gaussian observation noise, least squares and maximum likelihood have the same minimizer.
- Several common supervised losses are negative log-likelihoods for explicit observation models.

With only three observations, the Bernoulli MLE can lie on the boundary and report a zero probability for an unobserved outcome.

Three flips, three heads

You flip a coin from your pocket three times. Heads, heads, heads. MLE does what it always does:

$$\hat{\theta}_{\text{MLE}} = \frac{n_H}{n} = \frac{3}{3} = 1.0$$



The plug-in model at $\hat{\theta} = 1$ assigns probability zero to every future tail.

Consequence of the boundary estimate

$\hat{\theta} = 1$ is a checkable claim: it prices “at least one tail in the next 100 flips” at **exactly zero**.

- Background knowledge about ordinary coins makes this plug-in prediction implausible.
- The three flips are real evidence — but they are **three flips**.
- A Bayesian model represents such pre-data information with a prior distribution.

MLE has no slot for what you knew before the data. Today we build that slot: the **prior.**

Zero-count estimates in simple models

System	The three-heads bug, at work
spam filter	“invoice” never appeared in the spam folder → $P(\text{invoice} \mid \text{spam}) = 0 \rightarrow$ one word acquits any email, forever
drug trial	3 successes in 3 patients → “this drug <i>cannot</i> fail”
language model	a letter pair never seen in training → probability 0 → $\log 0 = -\infty$ loss on any text containing it

L26 applies the corresponding Dirichlet update to character counts, so unseen transitions receive nonzero posterior-predictive probability.

What would a fix even look like?

Three demands, straight from your intuition:

1. Tiny datasets must **not** produce absolute certainty.
2. For a fixed prior that assigns positive density near the data-generating parameter, its influence should diminish as evidence accumulates.
3. The hunch itself must be **adjustable** — a fairness zealot and a con-artist-spotter should be able to encode different starting beliefs.

Today's route: Bayes' rule defines the posterior; MAP extracts its mode; conjugacy gives closed-form posterior updates.

Learning outcomes

By the end of this lecture you will be able to:

1. Read Bayes' rule as a **belief update**, naming all four factors and their jobs.
2. Compute **MAP** estimates and say exactly when $\text{MAP} = \text{MLE}$.
3. Use the **Beta distribution** as a prior over probabilities, reading α, β as pseudo-counts.
4. Derive ★ the **Beta–Bernoulli conjugate update**, $\text{Beta}(\alpha + n_H, \beta + n_T)$.
5. Update **sequentially** as data streams in — and explain why order can't matter.
6. State the **Gaussian–Gaussian** posterior as a precision-weighted average.
7. Interpret **L2 regularization** as Gaussian-prior MAP (and L1 as Laplace-prior MAP).

Bayes' rule: the belief-update machine

The four factors, named

One rule merges what you believed with what you saw:

$$p(\theta \mid D) = \frac{P(D \mid \theta) \cdot p(\theta)}{P(D)}$$

Factor	Name	Its job
$p(\theta \mid D)$	posterior	density of θ <i>after</i> the data
$P(D \mid \theta)$	likelihood	how loudly the data argues for each θ — L14's object
$p(\theta)$	prior	density of θ <i>before</i> the data
$P(D)$	evidence	one number that makes the left side a distribution

Numbers first: a bag of coins

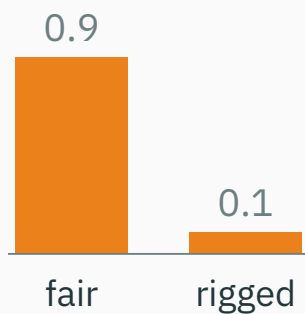
A bag holds 10 coins: **9 fair** ($\theta = 0.5$), **1 rigged** ($\theta = 0.9$). You draw one at random, flip it 3 times: **3 heads**. Did you draw the rigged one?

Two hypotheses $\rightarrow$ Bayes' rule is a two-row table:

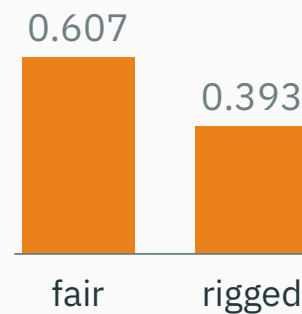
Coin	prior	likelihood θ^3	prior $\times$ lik	posterior
fair	0.9	$0.5^3 = 0.125$	0.1125	$0.1125/0.1854 = \mathbf{0.607}$
rigged	0.1	$0.9^3 = 0.729$	0.0729	$0.0729/0.1854 = \mathbf{0.393}$

Three heads moved the posterior probability of “rigged” from **10% to 39%**. The update multiplies prior by likelihood and rescales the two numbers to sum to 1.

before: **prior**



after 3 heads: **posterior**



```
prior = np.array([0.9, 0.1])      # P(fair), P(rigged)
like  = np.array([0.5**3, 0.9**3]) # P(HHH | each coin)
post  = prior * like              # multiply...
post /= post.sum()               # ...normalize -> [0.607, 0.393]
```

This finite-hypothesis example implements Bayes' rule by multiplication and normalization.

The evidence is bookkeeping

For discrete hypotheses, $P(D) = \sum_h P(D | h)P(h)$. For a continuous parameter, $P(D) = \int P(D | \theta)p(\theta) d\theta$. In either case it is constant with respect to θ and cannot move the posterior mode.

$$p(\theta | D) \propto P(D | \theta) \cdot p(\theta)$$

Posterior $\propto$ likelihood $\times$ prior. The evidence is the normalization constant that makes the posterior integrate to one.

The proportional form is sufficient whenever only the posterior shape or mode is needed.

MAP: maximize the posterior density

Same recipe as MLE — but climb the **posterior** instead of the likelihood:

$$\theta_{\text{MLE}} = \arg \max_{\theta} P(D \mid \theta) \qquad \theta_{\text{MAP}} = \arg \max_{\theta} P(D \mid \theta) \cdot p(\theta)$$

In log-space (products $\rightarrow$ sums — L2's rule, as always):

$$\theta_{\text{MAP}} = \arg \max_{\theta} \left[\underbrace{\log P(D \mid \theta)}_{\text{log-likelihood}} + \underbrace{\log p(\theta)}_{\text{log-prior density}} \right]$$

MAP = maximum a posteriori. On the log scale, MAP maximizes the sum of log-likelihood and log-prior density.

A flat prior changes nothing

Take $p(\theta) = 1$ on $[0, 1]$. Then $\log p(\theta) = 0$:

$$\theta_{\text{MAP}} = \arg \max_{\theta} [\log P(D \mid \theta) + 0] = \theta_{\text{MLE}}$$

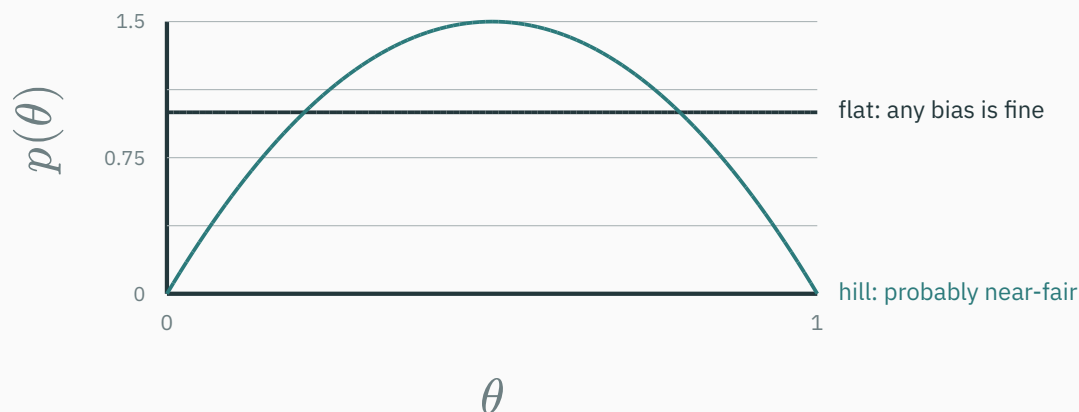
In this bounded parameterization, a uniform prior makes the MAP and MLE optimizers coincide. This does not mean that MLE assumes a prior: MLE and MAP are distinct procedures, and a density uniform in one parameterization need not be uniform after reparameterization.

A prior changes the posterior and MAP estimate most visibly in small samples. For fixed regular priors with support near the data-generating parameter, its influence on the MAP estimate typically diminishes as n grows.

The coin gets a prior

A prior must live on $[0, 1]$

θ is a probability, so a prior is a **density on $[0, 1]$** (L12: continuous belief is area).
Two candidates:



The hill is $p(\theta) \propto \theta(1 - \theta)$ — zero at the impossible extremes, gentle peak at $1/2$. (It has a family name; we'll meet the family in a few slides.)

Worked: the flat prior

Multiply the likelihood of three heads by the flat prior:

$$p(\theta \mid D) \propto \underbrace{\theta^3}_{\text{likelihood}} \cdot \underbrace{1}_{\text{prior}} = \theta^3$$

Climbing θ^3 on $[0, 1]$ ends at the right wall:

$$\theta_{\text{MAP}} = 1.0 = \theta_{\text{MLE}}$$

The boundary estimate survives a flat prior. Moving it requires an informative prior.

Worked: the hill prior, by hand

Same three heads, hill prior $p(\theta) \propto \theta(1 - \theta)$:

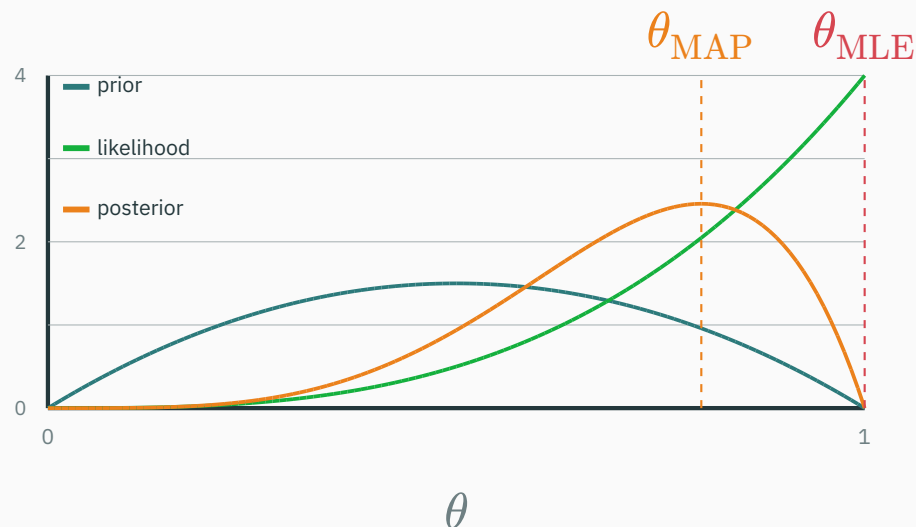
$$p(\theta \mid D) \propto \theta^3 \cdot \theta(1 - \theta) = \theta^4(1 - \theta)$$

Maximize with one derivative (drop constants; they can't move a peak):

$$\frac{d}{d\theta} [\theta^4(1 - \theta)] = 4\theta^3 - 5\theta^4 = \theta^3(4 - 5\theta) = 0$$

$$\Rightarrow \theta_{\text{MAP}} = \frac{4}{5} = 0.8$$

Same three heads: the flat prior says 1.00; a gentle near-fair belief says 0.80.



Likelihood rescaled to area 1 so the three shapes share one axis.

The posterior mode lies between the prior mode (0.5) and likelihood maximum (1.0), with the relative concentration of the two factors determining its position.

Check it by computer

No calculus, just a grid — the same argmax, numerically:

```
th    = np.linspace(0, 1, 1001)
flat  = th**3                # likelihood x flat prior
hill  = th**3 * th * (1 - th) # likelihood x hill prior

th[flat.argmax()], th[hill.argmax()]    # -> (1.0, 0.8)
```

- Shapes only — no normalizing constants anywhere, because `argmax` ignores scale.
- This grid trick works for **any** prior you can evaluate — calculus optional. (T7 uses it to cross-check every derivation.)

A coin shows 9 heads in 10 flips. With a **flat** prior on θ , the MAP estimate is...

A. 0.50 — the prior always pulls to fair

B. 0.90 — exactly the MLE

C. 0.83 — pulled part-way toward 1/2

D. 1.00 — MAP always sharpens the MLE

Correct: B — $\theta_{\text{MAP}} = 0.90 = \theta_{\text{MLE}}$

Why: A flat prior adds $\log 1 = 0$ to the log-posterior — the argmax cannot move, so $\text{MAP} = \text{MLE} = 9/10$. Answer C (≈ 0.83) is what the *hill* prior $\text{Beta}(2, 2)$ would give — right instinct, wrong prior; A forgets the data entirely; D isn't a thing MAP does.

The Beta family as a prior over probabilities

The hill has a family name

Tilt the exponents and the hill becomes a whole family of beliefs on $[0, 1]$:

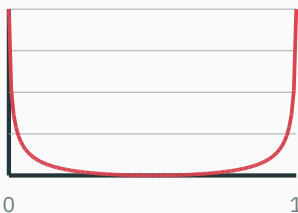
$$\text{Beta}(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- $\Gamma(k) = (k-1)!$ for positive integers, and the gamma function extends this definition so α and β need not be integers.
- Our hill $\propto \theta(1-\theta)$ is $\text{Beta}(2, 2)$; the flat prior is $\text{Beta}(1, 1)$.

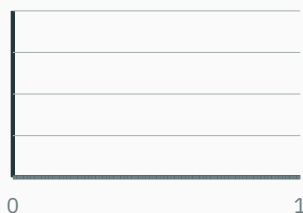
$B(\alpha, \beta)$ exists so the area is 1. We will **never** integrate it in this lecture — inspection will hand it to us for free.

All dials from $\{0.5, 1, 2, 5\}$ — one family, five personalities:

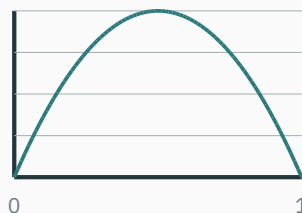
Beta(0.5, 0.5) — extremist



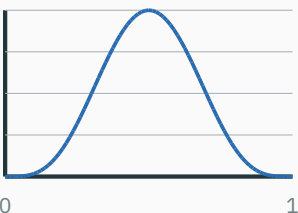
Beta(1, 1) — agnostic



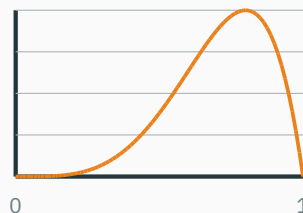
Beta(2, 2) — gentle hunch



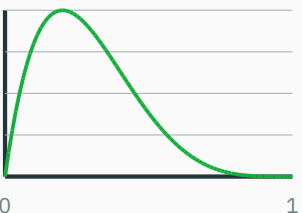
Beta(5, 5) — firmer hunch



Beta(5, 2) — leans heads



Beta(2, 5) — leans tails



One knob pair covers superstition, indifference, hunches, conviction, and lopsided suspicion.

A mode-based pseudo-count mnemonic

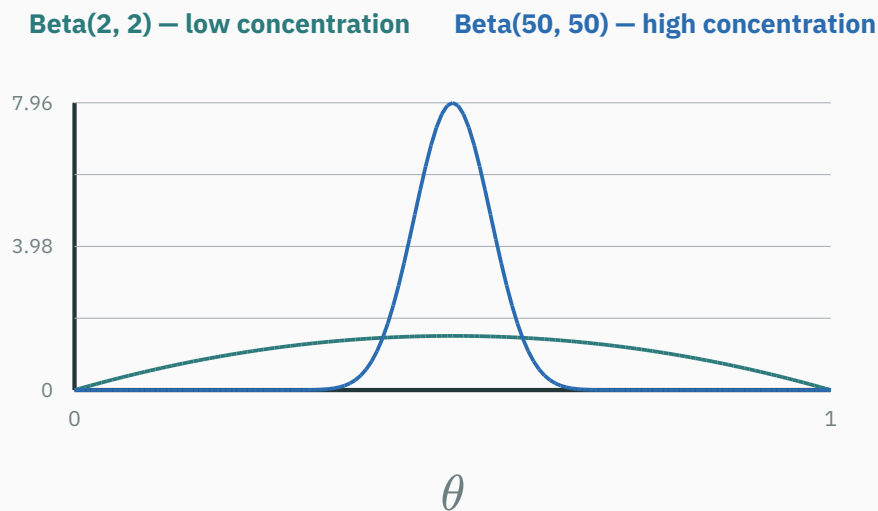
For $\alpha, \beta > 1$ the peak sits at $\text{mode} = (\alpha - 1)/(\alpha + \beta - 2)$ — exactly the MLE of a record with $\alpha - 1$ heads and $\beta - 1$ tails.

Prior	Imaginary record	Peak
Beta(1, 1)	0 H, 0 T — no memory at all	flat
Beta(2, 2)	1 H, 1 T — the gentlest fairness hint	0.5
Beta(5, 2)	4 H, 1 T — suspects a heads bias	0.8
Beta(50, 50)	49 H, 49 T — fairness conviction	0.5

For MAP, $\alpha - 1$ and $\beta - 1$ act like added head and tail counts. This is a mode mnemonic; the posterior update itself adds observations directly to α and β .

Concentration is controlled by $\alpha + \beta$

Same center, different bankroll: mean = $\alpha/(\alpha + \beta)$ fixes *where*; the total $\alpha + \beta$ fixes *how firmly*.



A Beta(50, 50) prior is much more concentrated than Beta(2, 2), so the same dataset moves it less.

Play the update before we derive it

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/bayesian-posterior

Drag the prior's α, β , stream in evidence across 8 experiments, and watch $\text{prior} \times \text{likelihood} = \text{posterior}$ re-render live.

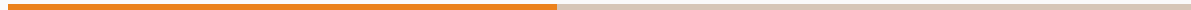
INTERACTIVE

↗ seeing-theory.brown.edu/bayesian-inference/index.html

Seeing Theory · Bayesian Inference — flip a coin one toss at a time and watch a Beta posterior fill in.

Both toys are running one identity — the one we now derive by hand.

Conjugacy: the one-line update



Set up the multiplication

Data: n_H heads, n_T tails ($n = n_H + n_T$). L14's likelihood, and a $\text{Beta}(\alpha, \beta)$ prior — constants dropped, shapes kept ($\propto$ absorbs $1/B$):

$$p(\theta \mid D) \propto \underbrace{\theta^{n_H} (1 - \theta)^{n_T}}_{\text{likelihood}} \cdot \underbrace{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}_{\text{prior density}}$$

Both factors speak the **same language**: powers of θ and powers of $(1 - \theta)$. That coincidence is about to do all the work.

Same base, so multiplication is addition upstairs:

$$p(\theta \mid D) \propto \theta^{n_H + \alpha - 1} (1 - \theta)^{n_T + \beta - 1}$$

Stare at it: this is **the Beta shape again**, with new dials

$$\alpha' = \alpha + n_H, \quad \beta' = \beta + n_T$$

The head and tail counts add to the two exponents. The data do not change the family, only its parameters.

We hold a function $\propto \theta^{\alpha'-1}(1-\theta)^{\beta'-1}$ that must integrate to 1 (it's a posterior). Among all rescalings $c \cdot \theta^{\alpha'-1}(1-\theta)^{\beta'-1}$, **exactly one** has area 1 — and $\text{Beta}(\alpha', \beta')$ is that one, by definition.

So the constant is forced, no integral computed:

$$\text{Beta}(\alpha, \beta) \text{ prior} + n_H \text{ heads}, n_T \text{ tails} \Rightarrow \text{posterior} = \text{Beta}(\alpha + n_H, \beta + n_T)$$

“By inspection” means recognizing the Beta kernel and using its known normalizing constant.

Conjugacy, named

Definition. A prior family is **conjugate** to a likelihood if the posterior always lands back in the same family.

- Beta in $\rightarrow$ Bernoulli data $\rightarrow$ Beta out: the Beta family is conjugate to coin flips.
- The entire update is **two integer additions**: $\alpha \leftarrow \alpha + n_H$, $\beta \leftarrow \beta + n_T$. No integral, no grid, no optimizer.
- Our worked example: Beta(2, 2) + 3 heads $\rightarrow$ Beta(5, 2), whose mode is 4/5.

The MAP formula drops out

The posterior is $\text{Beta}(\alpha + n_H, \beta + n_T)$, and we know where a Beta peaks:

$$\theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{n + \alpha + \beta - 2}$$

- Check: 3 heads, $\text{Beta}(2, 2)$: $(3 + 1)/(3 + 2) = 4/5 = 0.8$ — matches the hand derivative. ✓
- Check: flat prior $\alpha = \beta = 1$: $\theta_{\text{MAP}} = n_H/n = \theta_{\text{MLE}}$. ✓

The MAP estimate is the empirical count formula with the mode offsets $\alpha - 1$ and $\beta - 1$.

Stress test: 9 heads, 1 tail

Ten flips, nine heads ($\theta_{\text{MLE}} = 0.9$). Three believers meet the same data:

Prior	θ_{MAP}	Reading
Beta(1, 1) — agnostic	$9/10 = 0.900$	no opinion → data speaks alone
Beta(2, 2) — gentle hunch	$10/12 \approx 0.833$	two pseudo-flips nudge back toward fair
Beta(50, 50) — conviction	$58/108 \approx 0.537$	98 pseudo-flips: ten real ones barely dent it

Same data, three defensible answers — the prior is a **published, criticizable assumption**, not a hidden thumb on the scale. (Which believer is being unreasonable? That argument is now about α, β — in the open.)

The conjugate catalog

The same algebra appears beyond coins:

Likelihood	Conjugate prior	Posterior update
Bernoulli (coin)	Beta	$\alpha \leftarrow \alpha + n_H, \beta \leftarrow \beta + n_T$
Categorical (d faces)	Dirichlet	add each face's count to its pseudo-count
Gaussian mean (σ known)	Gaussian	precision-weighted average — next section

Continuity ledger — planted today, paid off in L26: the Dirichlet row is Beta's d -sided sibling. When L26's character language model meets an unseen letter pair, *Dirichlet smoothing* (add- α pseudo-counts) is exactly this table's second row.

Prior Beta(2, 3); you then observe 4 heads and 1 tail. The posterior is...

A. Beta(6, 4)

B. Beta(3, 7)

C. Beta(5, 2)

D. Beta(7, 8)

Correct: A — **Beta**(2 + 4, 3 + 1) = **Beta**(6, 4)

Why: Heads land on α , tails on β — nothing else moves. B swapped the two counts; C threw the prior away and updated a flat Beta(1, 1); D added the total $n = 5$ to both dials, double-counting every flip.

Watching beliefs sharpen

Today's posterior is tomorrow's prior

Nothing forces the flips to arrive in one batch. Feed them one at a time, and **each posterior becomes the next prior**:

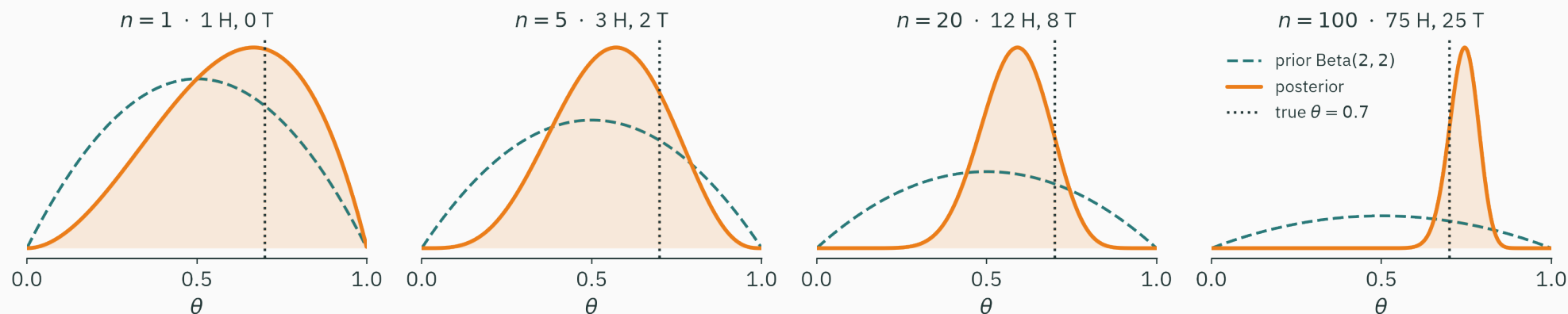
$$\text{Beta}(\alpha, \beta) \xrightarrow{\text{H}} \text{Beta}(\alpha + 1, \beta) \xrightarrow{\text{T}} \text{Beta}(\alpha + 1, \beta + 1) \xrightarrow{\text{H}} \dots$$

```
a, b = 2, 2          # Beta(2,2): 1 imaginary head, 1 tail
for y in flips:      # y = 1 for heads, 0 for tails
    a, b = a + y, b + 1 - y  # the ENTIRE Bayesian update
# at any moment: posterior = Beta(a, b), MAP = (a-1)/(a+b-2)
```

Two integers of memory, one addition per observation. This is **online learning** in its purest form — the model never re-reads old data.

One hundred flips, four snapshots

A coin with true $\theta = 0.7$ streams flips into a Beta(2, 2) prior:



After 1 flip the prior still dominates; by 100 flips (75 H, 25 T $\rightarrow$ Beta(77, 27)) the posterior is a needle — parked at the **sample's** rate 0.75, a whisker above the truth. That residual wobble is the data's own sampling noise, not the prior's doing.

The posterior sharpens like $1/\sqrt{n}$

How fast is certainty earned? The posterior's spread obeys

$$\text{sd} \approx \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}}$$

- Quadruple the flips → **halve** the uncertainty — the same $\sqrt{n}$ law you met in L13's CLT demo, now running *inside* the posterior.
- By $n = 100$, the prior's 4 pseudo-flips are 4 votes in an electorate of 104: present, out-voted, and fading.

Order cannot matter

Flip H then T, starting from $\text{Beta}(\alpha, \beta)$: $(\alpha + 1, \beta)$, then $(\alpha + 1, \beta + 1)$. Flip T then H: $(\alpha, \beta + 1)$, then $(\alpha + 1, \beta + 1)$. **Same place.**

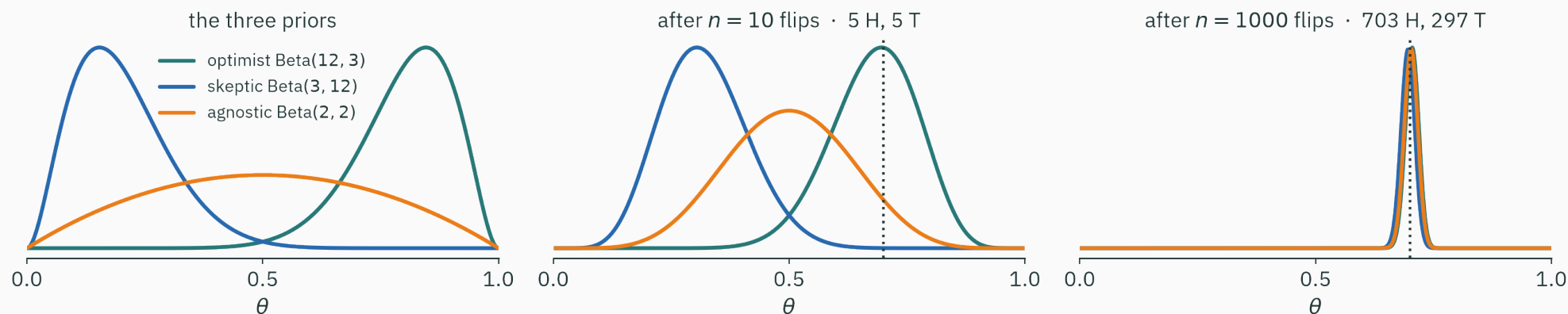
No accident: the posterior is a **product** of per-flip factors, and multiplication commutes —

$$p(\theta \mid D) \propto p(\theta) \prod_{i=1}^n P(y_i \mid \theta)$$

Stream the data or dump it in one batch, in any order — **exactly the same posterior.**

Three scientists, three priors

An optimist, a skeptic, and an agnostic watch the same coin ($\theta = 0.7$):



Ten flips in (5 H, 5 T), the posteriors still differ. After a thousand flips they are nearly indistinguishable.

When the prior washes out

The formula says exactly when belief stops mattering:

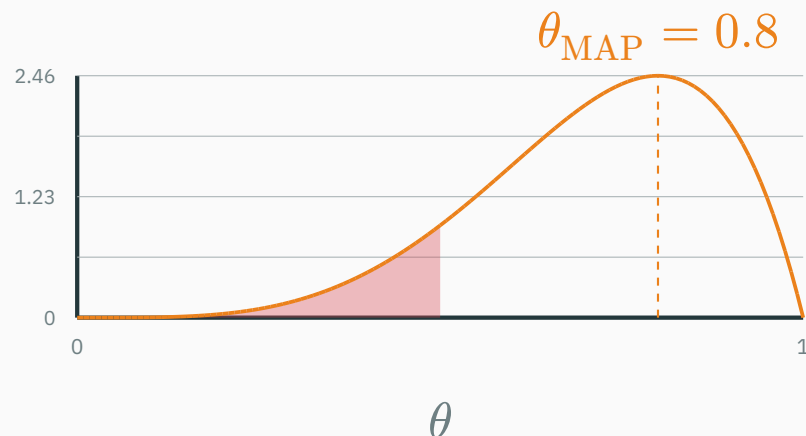
$$\theta_{\text{MAP}} = \frac{n_H + \alpha - 1}{n + \alpha + \beta - 2} \rightarrow \frac{n_H}{n} = \theta_{\text{MLE}} \quad \text{as } n \rightarrow \infty$$

- The prior is a **fixed** number of pseudo-votes in a growing electorate: decisive at $n = 3$, a rounding error at $n = 10^6$.
- With a fixed regular prior and a growing i.i.d. sample, MLE and MAP typically converge to the same value.

For this fixed Beta prior, the difference between MAP and MLE is of order $1/n$. Priors with zero support near the truth or priors that change with n need not wash out this way.

Posterior inference for the opening coin example

Three heads in three flips, Beta(2, 2) prior $\rightarrow$ posterior Beta(5, 2):



- The MAP estimate is $\theta_{\text{MAP}} = 0.8$, while the posterior still assigns about 11% probability to $\theta < 0.5$.
- After three flips, the posterior remains uncertain and assigns nonzero probability to future tails.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mle-map-coin

A coin you control: set the true bias, choose α, β , and flip. Watch θ_{MAP} glide from the prior's peak toward θ_{MLE} as the pseudo-counts get out-voted.

Try a strong prior far from the data, then compare it with a flat prior at $n = 3$.

Gaussian meets Gaussian

A continuous analogue

New estimation problem: a sensor reads a true value μ through Gaussian noise (L13):

$$x_i \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma \text{ known}, \quad i = 1, \dots, n$$

You also have a hunch — the calibration sheet says readings sit near μ_0 :

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$$

Beta was the prior for a **probability**; the Gaussian is the prior for a **location**. And the Gaussian is conjugate to itself: Gaussian in $\rightarrow$ Gaussian out.

Stated: Gaussian in, Gaussian out

Multiply the two bells and complete the square (algebra in T7 and MML 8.3.2 — today, the shape of the answer). Call $1/\text{variance}$ **precision** — confidence, as a number: prior precision $\tau_0 = 1/\sigma_0^2$, data precision $\tau_D = n/\sigma^2$.

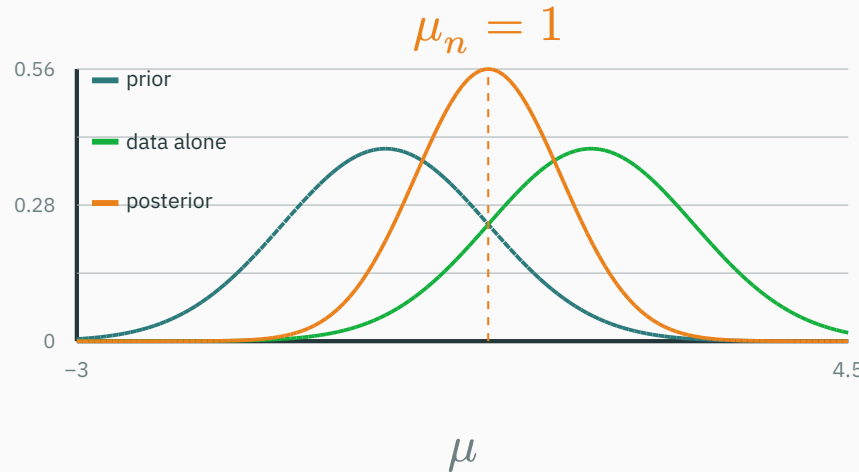
$$p(\mu \mid D) = \mathcal{N}(\mu_n, \sigma_n^2), \quad \frac{1}{\sigma_n^2} = \tau_0 + \tau_D$$

$$\mu_n = \frac{\tau_0 \mu_0 + \tau_D \bar{x}}{\tau_0 + \tau_D}$$

Precisions add, and the posterior mean is a **precision-weighted average** of hunch μ_0 and data average $\bar{x}$ — the surer voice counts more.

Hunch $\mu_0 = 0$ (precision $1/\sigma_0^2 = 1$); data $n = 4$, $\bar{x} = 2$, $\sigma^2 = 4$ (precision $n/\sigma^2 = 1$).
Equal confidence $\rightarrow$ meet exactly halfway:

$$\mu_n = \frac{1 \cdot 0 + 1 \cdot 2}{1 + 1} = 1, \quad \sigma_n^2 = \frac{1}{1 + 1} = 0.5$$



The posterior is **narrower than both parents** — two independent confidences stacked.

Precision is the currency

- Posterior mean = average of the two centers, **weighted by precision** — the surer voice counts more.
- The prior behaves like σ^2/σ_0^2 **virtual observations** pinned at μ_0 — pseudo-counts again, in continuous clothing.
- As n grows, data precision $n/\sigma^2 \rightarrow \infty$, so a fixed finite prior precision has diminishing influence.

**Conjugate updating is an average where precision is the currency:
pseudo-flips for coins, virtual readings for means.**

MAP interpretations of common penalties

Recall the Gaussian-noise result from L14

Linear regression, Gaussian noise: $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. L14 showed:

$$\text{NLL}(\mathbf{w}) = \frac{1}{2\sigma^2} \|\mathbf{y} - X\mathbf{w}\|^2 + \text{const}$$

Under independent Gaussian observation noise with fixed variance, minimizing NLL is equivalent to least squares.

To form a MAP estimator, place an explicit prior on the weights.

Give the weights a prior

Belief: weights shouldn't be huge — a model that needs $w = 10^6$ is probably memorizing noise. Encode it:

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 I) \Rightarrow -\log p(\mathbf{w}) = \frac{1}{2\sigma_0^2} \|\mathbf{w}\|^2 + \text{const}$$

MAP swaps argmax-posterior for argmin of the negative log-posterior:

$$\mathbf{w}_{\text{MAP}} = \arg \min_{\mathbf{w}} \underbrace{\frac{1}{2\sigma^2} \|\mathbf{y} - X\mathbf{w}\|^2}_{\text{data misfit (L14)}} + \underbrace{\frac{1}{2\sigma_0^2} \|\mathbf{w}\|^2}_{\text{prior's fine on big weights}}$$

```
nll = ((y - X @ w)**2).sum() / (2 * sig2)    # L14: Gaussian NLL
nlp = (w**2).sum() / (2 * sig0_2)            # -log N(0, sig0^2) prior
loss = nll + nlp                             # MAP objective
```


MAP with a Gaussian prior is ridge regression

Multiply through by $2\sigma^2$ (argmins don't care):

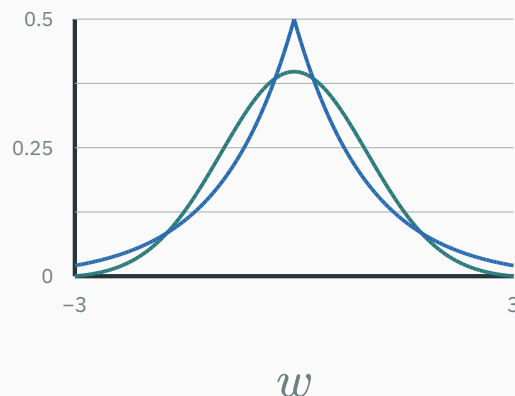
$$\mathbf{w}_{\text{MAP}} = \arg \min_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|^2 + \lambda \|\mathbf{w}\|^2, \quad \lambda = \frac{\sigma^2}{\sigma_0^2}$$

Ridge regression is MAP with a Gaussian prior on the weights. The coefficient λ is the noise-to-prior-variance ratio.

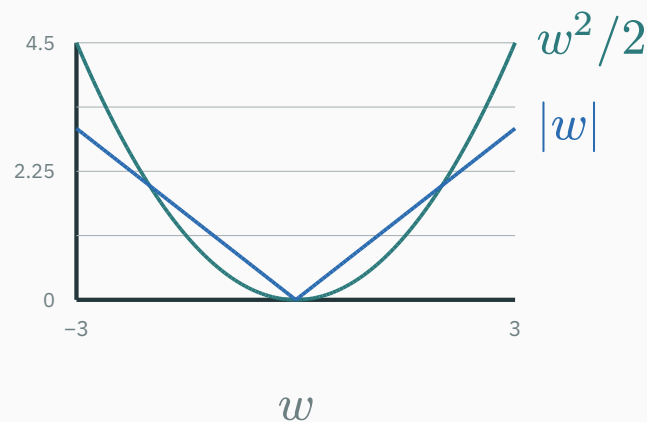
A later ML course may introduce ridge regression or weight decay as regularization, with λ tuned by validation. In this probabilistic model, increasing λ corresponds to decreasing the prior variance σ_0^2 and concentrating more prior mass near zero.

A prior density contributes its negative logarithm to the MAP objective:

the prior — Gaussian vs Laplace



the penalty — $\log p(w)$



Gaussian prior $\rightarrow$ quadratic penalty $\rightarrow$ **L2 / ridge**. Laplace prior $\rightarrow |w|$ penalty $\rightarrow$ **L1 / lasso**; the corner at zero allows exact zero-valued coefficients.

The regularization dictionary

You add to the loss...	You just assumed...	Its street name
$\lambda \ w\ ^2$	Gaussian prior $\mathcal{N}(0, \sigma_0^2)$	ridge / weight decay
$\lambda \ w\ _1$	Laplace prior	lasso
add- α pseudo-counts	Beta / Dirichlet prior	Laplace smoothing ($\rightarrow$ L26)

If $\exp(-R(w))$ is integrable, a penalty R can be interpreted as a negative log-prior in MAP estimation. Some training regularizers and constraints do not admit this simple interpretation.

Summary

Lecture 15 — summary

- **Bayes**: posterior $\propto$ likelihood $\times$ prior — the evidence only normalizes.
- **MAP** climbs log-likelihood + log-prior; flat prior $\Rightarrow$ MAP = MLE.
- **Beta**(α, β): the MAP mode has offsets $\alpha - 1, \beta - 1$; $\alpha + \beta$ controls concentration.
- **Conjugacy** ★: Beta prior + coin data $\rightarrow$ Beta($\alpha + n_H, \beta + n_T$) — two additions.
- **Sequential = batch**, any order; width $\sim 1/\sqrt{n}$; data swamps the prior.
- **Gaussians**: precisions add; posterior mean = precision-weighted average.
- **MAP penalty interpretations**: Gaussian $\rightarrow$ ridge, Laplace $\rightarrow$ lasso, Beta/Dirichlet $\rightarrow$ smoothing.

Read before L16 — MML 6.6.1 & 8.3.2 · MacKay Ch. 3. **T7** pairs with this lecture (grid-check every derivation); **Quiz 3 / midsem** covers Module 3.

Practice problems

Try on paper; verify with the T7 notebook's grid trick.

1. Re-derive the ★ update: $\text{Beta}(\alpha, \beta)$ prior, n_H heads, n_T tails $\rightarrow$ posterior, from a blank page.
2. Prior $\text{Beta}(5, 5)$, data 2 H, 8 T: posterior, θ_{MAP} , posterior mean — and compare with θ_{MLE} .
3. Show that $\text{Beta}(1, 1)$ makes $\theta_{\text{MAP}} = \theta_{\text{MLE}}$ for **every** possible dataset.
4. How many **consecutive heads** push θ_{MAP} above 0.99 under a $\text{Beta}(2, 2)$ prior? (Solve $(n + 1)/(n + 2) > 0.99$.)
5. Sensor: prior $\mathcal{N}(0, 1)$; data $n = 9$, $\bar{x} = 3$, $\sigma^2 = 9$. Posterior mean and variance?
6. A colleague doubles ridge's λ . What happens to σ_0^2 in the MAP interpretation, and which prior would produce an L1 penalty instead?



MAP keeps the peak, throws the width

★ OPTIONAL

MAP reports **one number** — but the posterior was a whole curve, and the curve knows more.

- Predicting the next flip by **averaging over the posterior** gives the posterior *mean*, not the mode: $P(\text{next H} \mid D) = \frac{\alpha + n_H}{\alpha + \beta + n}$.
- Three heads, Beta(2, 2): predict $5/7 \approx 0.71$ — milder than $\theta_{\text{MAP}} = 0.8$, and a world away from MLE's 1.0.
- Keeping the **entire** posterior (and integrating over it) is Bayesian machine learning proper — the Probabilistic ML course lives there.



Laplace's rule of succession

★ OPTIONAL

Flat prior + posterior mean $\rightarrow P(\text{next H}) = \frac{n_H + 1}{n + 2}$ — “add one imaginary head and one imaginary tail.”

- Laplace, 1774: what is the probability the sun rises tomorrow, given n recorded sunrises? Answer: $(n + 1)/(n + 2)$ — huge, but **never 1**.
- NLP's **add-one smoothing** is this rule, verbatim, applied to word counts — and its add- α generalization is the Dirichlet machinery L26 uses to keep the language model's $\log 0$ at bay.

250 years old, still shipping in production systems. Priors age well.

L2 and L1 penalties are negative log-priors in MAP estimation.

Next: **Module 4 — The Optimization Landscape**. We now know *what* to minimize; L16 begins the story of *how*.